COMP4423 – Computer Vision

Assignment 2

PolyU Main Campus Image Classification

[Deadline: 23:59 HKT on Wednesday, 28 October 2026]

1. Task Description

In this assignment, you will build an end-to-end image classification pipeline using a self-collected dataset. Given a scene photograph, your program should predict whether it was taken inside or outside the PolyU main campus. You will collect and label images, extract visual features, train and evaluate a traditional machine learning classifier, and build a simple local application for image upload or file selection. The objective is to become familiar with the standard machine learning paradigm through a complete, reproducible real-world task.

Class definitions and labeling rules

Class label	Definition
inside (1)	A photograph taken within the PolyU main campus, showing a scene within that campus.
outside (0)	A photograph taken outside the PolyU main campus, showing a scene outside that campus.

Geographic scope. Use PolyU's official Campus Map as the geographic reference and record where each photograph was taken. Only the main campus is the positive class; off-campus premises do not become "inside" merely because they are affiliated with PolyU. "Inside/outside campus" does not mean "indoors/outdoors." Reference: PolyU Campus Map

Unambiguous scenes. The camera position and the main depicted scene must be clearly on the same side of the campus boundary. Exclude uncertain boundary locations, gate thresholds, connecting bridges, and views looking across the boundary. Do not label images by red bricks, logos, or appearance alone.

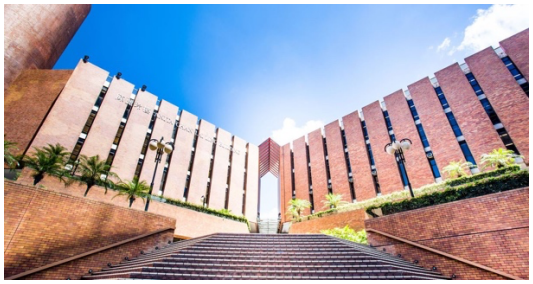
Visual input only. Collect scene-level images with useful surroundings, not isolated objects or photographs of other images. Location notes and GPS may help establish ground truth, but the classifier must use image pixels only, not GPS/EXIF, filenames, folder names, or external location lookups.

Example Images

The following photographs illustrate the class labels and the location-group rule. They are reference material only; collect your own dataset for Task 2.

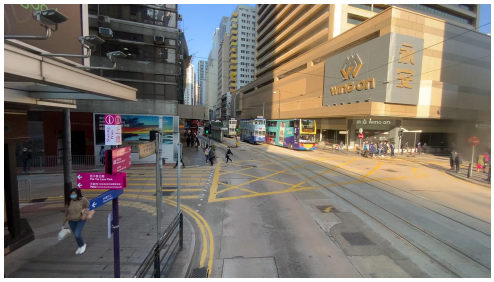
Figure 1. Examples of the two classes

Inside PolyU main campus



Main-campus courtyard near Anita Chan Lai Ling Building.

Outside PolyU main campus



Public street beside Wing On Centre, Sheung Wan.

Labels describe geographic location, not whether the scene is outdoors or whether an image is in color. Record the collection location and apply the boundary rules on the previous page.

2. Tasks & Assessment

Total: 100 marks.

Task 1: Class definition & data collection plan. (5 marks)

Apply the two class definitions above and identify your planned collection locations on an annotated map or location list. Explain how you will vary viewpoints, distances, lighting, and backgrounds. Include comparable scene types in both classes, such as buildings, walkways, and green or open spaces. Include some outside scenes that look similar to campus scenes, rather than collecting only easy negatives.

Task 2: Dataset building & labeling. (10 marks)

Collect your own photographs and organize them using the class labels inside and outside. There is no fixed requirement for the number of images or location groups. Build a reasonable, diverse dataset sufficient for model training and meaningful evaluation, and briefly justify its scale in your report.

Item	Requirement
Location diversity	Cover varied locations in both classes. Nearby views of the same courtyard, building, or landmark belong to one location group.
Data split	Split location groups separately within each class, aiming for 60% training, 20% validation, and 20% test. Ensure both classes are represented in each split.

Prevent leakage. Keep all views of the same scene, including repeat visits, burst/video frames, and edited copies, in the same location group and split. Do not randomly split individual images from one scene across sets. Apply augmentation only to training images after splitting; report actual image and location counts for every class and split.

Dataset records. Photograph only where permitted; avoid identifiable people, personal information, and restricted areas.

Task 3*: Train a classifier. (10 marks)

Train a classifier to identify your location classes using a traditional machine learning pipeline (e.g., Random Forest, SVM, Gradient Boosting, etc.). Deep learning approaches (such as neural networks) are explicitly excluded for this task. You must justify your key algorithmic choices, provide training evidence, and ensure your pipeline is fully reproducible with your code.

Task 4*: Evaluation & error analysis. (5 marks)

Evaluate on a held-out test set and report accuracy, confusion matrix, per-class metrics, and so on. Include qualitative examples of correct and incorrect predictions, and discuss typical failure modes (e.g., background bias, lighting shifts, similar-looking location, viewpoint changes). Specifically, analyze how these failure modes relate to the limitations of your chosen feature extraction methods.

Task 5: Application. (20 marks)

Build a simple local application that allows a user to upload or select an image, displays the image, and returns "Inside PolyU main campus" or "Outside PolyU main campus." Use the saved pipeline and exactly the same preprocessing as in evaluation; do not infer the answer from the file path or ask the user for the class. A desktop GUI or local web interface is sufficient; camera input and real-time processing are optional. The application must run locally and include brief instructions on how to launch and use it.

Task 6: Report. (50 marks)

A report to show the details of your method. Questions in the template should be answered.

Bonus: Submissions with excellent code quality, comment quality, output accuracy and report quality will be given a bonus of no more than 10 marks. The final grade of this assignment is min(100, normal grade + bonus).

Collaborate with Generative AI for problem-solving:** You are encouraged to use Generative AI to assist with implementation and debugging (e.g., dataset conversion, training scripts, evaluation or visualization utilities), but you must understand and verify the generated code and clearly describe in your report what AI helped with, what limitations or mistakes you observed, and how you corrected or improved the pipeline based on your own reasoning and testing. However, it's important to recognize that ***AI can make mistakes, so you should ***identify its limitations and areas for improvement*** in your report.

3. Submission

Follow the steps below:

1. Save the report as a PDF file and rename it, *e.g.*, `Assignment2_Report_12345678d_CHAN_Dawen.pdf`
2. Compress the implementation files (e.g., code, dataset^, etc.) into a .zip file and rename the .zip file. *e.g.*, `Assignment2_Code_12345678d_CHAN_Dawen.zip`
3. Upload the above **two files** to the Canvas system.

^Dataset submission (cloud-drive option)

You may include the dataset in the ZIP file or, if it is large, upload it to a cloud drive. For cloud submission, **include the sharing link in DATASET_LINK.txt within the ZIP file and in your report.** The link must allow **anyone with the link to view and download the complete dataset** without signing in, entering a password, requesting access, or needing special permissions. Test the link in a signed-out/private browser before submitting and keep it accessible until grading is complete.

Warning: If you are unable to complete the whole program, try to accomplish part of the tasks and make sure it can run successfully.

Any wrong file naming and submission will be given a ZERO mark in this assignment.

The submission deadline is **23:59 HKT on Wednesday, 28 October 2026**.

Late submission penalty: 10% is deducted for each day that the work is late. The penalty will be applied up to a maximum number of three days after and including the submission deadline day. After three days the work will be marked at zero.

This assignment is individual work. All work must be done on your own. Plagiarism is a serious offence. Copying code from web resources is prohibited as well. Any plagiarism case (for both the copier and the copiee) will be given a ZERO mark in this assignment.

4. Grading Criteria

Total: 100 marks

Tasks 1-4: Dataset, Training, and Evaluation (30 marks)

Criterion	Marks
Task 1: Apply the class definitions and explain a diverse data collection plan.	**5**
Task 2: Build a clearly labeled, organized dataset of sufficient scale and diversity.	**5**
Task 2: Split by location group, prevent leakage, and report class/split counts.	**5**
Task 3: Extract visual features and train a traditional classifier (no deep learning).	**5**
Task 3: Justify algorithm choices and provide training evidence and reproducible code.	**5**
Task 4: Report test metrics and prediction examples; relate errors to feature limitations.	**5**

Task 5: Application (20 marks)

Criterion	Marks
Upload or select an image and display it.	**5**
Load the saved classifier and use the same preprocessing as in evaluation.	**5**
Return the inside/outside prediction using image content only.	**5**
Run locally with clear launch and usage instructions.	**5**

Task 6: Report (50 marks)

Criterion	Marks
Problem description, motivation, and implementation details.	**10**
Experimental results and discussion.	**10**
Result analysis and reflection, including GenAI limitations where applicable.	**10**
Answers to all required template questions.	**10**
Clear organization, figures, tables, examples, and references.	**10**

Bonus: Up to 10 marks for exceptional code, comments, output, and report quality.
Final assignment score capped at 100.